

Reverse-check review package — Monotops_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	Monotops/model/Monotops_gen.fr
original model name	Monotops_gen (hidden from the agent)
paper	Monotops/text/1106.6199.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LMono (:=)

```
Module[{L0,L1,L120,L121,L120p,sp,sp1,sp2,f1,f2,c1,c2,c3,mu}, L0 := SMET
↪ (uqbar[sp1,f1,c1].uq[sp1,f2,c1] A0FC[f1,f2] +
↪ uqbar[sp1,f1,c1].Ga[5,sp1,sp2].uq[sp2,f2,c1] B0FC[f1,f2]); L1 := VMET[mu]
↪ (uqbar[sp1,f1,c1].Ga[mu,sp1,sp2].uq[sp2,f2,c1] A1FC[f1,f2] +
↪ uqbar[sp1,f1,c1].Ga[mu,sp1,sp2].Ga[5,sp2,sp].uq[sp,f2,c1] B1FC[f1,f2]); L120 := phiC[c3]
↪ Eps[c1,c2,c3] (CC[dqbar][sp1,f1,c1].dq[sp1,f2,c2] AQS[f1,f2] +
↪ CC[dqbar][sp1,f1,c1].Ga[5,sp1,sp2].dq[sp2,f2,c2] BQS[f1,f2]) + phiC[c1]
↪ (uqbar[sp1,f1,c1].FMET[sp1] A12S[f1] + uqbar[sp1,f1,c1].Ga[5,sp1,sp2].FMET[sp2]
↪ B12S[f1]); L120p := tphiC[c3] Eps[c1,c2,c3] (CC[dqbar][sp1,f1,c1].uq[sp1,f2,c2]
↪ tAQS[f1,f2] + CC[dqbar][sp1,f1,c1].Ga[5,sp1,sp2].uq[sp2,f2,c2] tBQS[f1,f2]) + tphiC[c1]
↪ (dqbar[sp1,f1,c1].FMET[sp1] tA12S[f1] + dqbar[sp1,f1,c1].Ga[5,sp1,sp2].FMET[sp2]
↪ tB12S[f1]); L121 := VC[mu,c3] Eps[c1,c2,c3]
↪ (CC[dqbar][sp1,f1,c1].Ga[mu,sp1,sp2].dq[sp2,f2,c2] AQV[f1,f2] +
↪ CC[dqbar][sp1,f1,c1].Ga[mu,sp1,sp2].Ga[5,sp2,sp].dq[sp,f2,c2] BQV[f1,f2]) + VC[mu,c1]
↪ (uqbar[sp1,f1,c1].Ga[mu,sp1,sp2].FMET[sp2] A12V[f1] +
↪ uqbar[sp1,f1,c1].Ga[mu,sp1,sp2].Ga[5,sp2,sp].FMET[sp] B12V[f1]); L0 + L1 + L120 + L121 +
↪ $Flag4F*L120p + HC[L0 + L1 + L120 + L121 + $Flag4F*L120p]]
```

Blank-slate reconstruction

Reconstructed Physics From sanitized.fr

Lagrangian

Conventions: u_i^a and d_i^a are SM up- and down-type Dirac quarks, with generation indices $i, j = 1, 2, 3$ and colour indices $a, b, c = 1, 2, 3$. Repeated indices are summed. Charge conjugation is written as

$$q^c \equiv C \bar{q}^T, \quad \bar{q}^c = (q^c)^\dagger \gamma^0.$$

The FeynRules object `CC[dqbar]` is therefore written as $\overline{d_i^{c,a}}$. The colour tensor `Eps[c1,c2,c3]` is ϵ_{abc} . The model adds the hermitian conjugate of all displayed interaction terms through `HC[...]`.

Let

$$\begin{aligned} S &\equiv \text{SMET}, & X_\mu &\equiv \text{VMET}, & \chi &\equiv \text{FMET}, \\ \phi^a &\equiv \text{phiC}, & \tilde{\phi}^a &\equiv \text{tphiC}, & V_\mu^a &\equiv \text{VC}. \end{aligned}$$

L0

$$\mathcal{L}_{\text{L0}} = S \bar{u}_i^a [A0FC_{ij} + B0FC_{ij}\gamma^5] u_j^a + \text{h.c.}$$

Equivalently, using chiral projectors,

$$A0FC_{ij} + B0FC_{ij}\gamma^5 = (A0FC_{ij} - B0FC_{ij})P_L + (A0FC_{ij} + B0FC_{ij})P_R.$$

L1

$$\mathcal{L}_{\text{L1}} = X_\mu \bar{u}_i^a \gamma^\mu [A1FC_{ij} + B1FC_{ij}\gamma^5] u_j^a + \text{h.c.}$$

Equivalently,

$$A1FC_{ij} + B1FC_{ij}\gamma^5 = (A1FC_{ij} - B1FC_{ij})P_L + (A1FC_{ij} + B1FC_{ij})P_R.$$

L120

$$\mathcal{L}_{\text{L120}} \supset \phi^c \epsilon_{abc} \overline{d_i^{c,a}} [AQ S_{ij} + BQ S_{ij}\gamma^5] d_j^b + \text{h.c.}$$

$$\mathcal{L}_{\text{L120}} \supset \phi^a \bar{u}_i^a [A12S_i + B12S_i\gamma^5] \chi + \text{h.c.}$$

L120p

Since $\text{\$Flag4F} = 1$, this term is active.

$$\mathcal{L}_{\text{L120p}} \supset \tilde{\phi}^c \epsilon_{abc} \overline{d_i^{c,a}} [tAQ S_{ij} + tBQ S_{ij}\gamma^5] u_j^b + \text{h.c.}$$

$$\mathcal{L}_{\text{L120p}} \supset \tilde{\phi}^a \bar{d}_i^a [tA12S_i + tB12S_i\gamma^5] \chi + \text{h.c.}$$

L121

$$\mathcal{L}_{\text{L121}} \supset V_\mu^c \epsilon_{abc} \overline{d_i^{c,a}} \gamma^\mu [AQ V_{ij} + BQ V_{ij}\gamma^5] d_j^b + \text{h.c.}$$

$$\mathcal{L}_{\text{L121}} \supset V_\mu^a \bar{u}_i^a \gamma^\mu [A12V_i + B12V_i\gamma^5] \chi + \text{h.c.}$$

Full LMono

$$\mathcal{L}_{\text{LMono}} = \mathcal{L}_{\text{L0}} + \mathcal{L}_{\text{L1}} + \mathcal{L}_{\text{L120}} + \mathcal{L}_{\text{L121}} + \mathcal{L}_{\text{L120p}}$$

The file contains no explicit $\text{DC}[\dots]$ covariant derivatives or $\text{FS}[\dots]$ field-strength terms. From the declared quantum numbers, all new states are $SU(2)_L$ singlets. With convention

$$D_\mu = \partial_\mu - ig_s G_\mu^A T_R^A - ig' Y B_\mu,$$

the colour-singlet neutral fields S , X_μ , and χ have $D_\mu = \partial_\mu$. The colour-triplet fields ϕ^a , $\tilde{\phi}^a$, and V_μ^a use the fundamental $SU(3)_c$ generators T^A , with hypercharges $Y = 2/3, -1/3$, and $2/3$, respectively. There is no $SU(2)_L$ gauge term.

Field Table

.fr symbol	Spin	$SU(3)_c$ rep	$SU(2)_L$ rep	$U(1)_Y$ / electric charge	Self- conjugate?	Mass
FMET	1/2	1	singlet	$Y = 0, Q = 0$	yes, Majorana- like	MFM = 50
VMET	1	1	singlet	$Y = 0, Q = 0$	yes, real vector	MVM = 50.
VC	1	3	singlet	$Y = 2/3, Q = 2/3$	no	MVC = 500
SMET	0	1	singlet	$Y = 0, Q = 0$	yes, real scalar	MSM = 50
phiC	0	3	singlet	$Y = 2/3, Q = 2/3$	no	MSC = 1000
tphiC	0	3	singlet	$Y = -1/3, Q = -1/3$	no	tMSC = 1000

Parameters

Parameter	Multiplies	Physical meaning	Default nonzero entries
A0FC[i, j]	$S \bar{u}_i u_j$	scalar flavour-changing up-quark coupling to SMET	$A0FC_{13} = A0FC_{31} = 0.1$
B0FC[i, j]	$S \bar{u}_i \gamma^5 u_j$	pseudoscalar up-quark coupling to SMET	all zero
A1FC[i, j]	$X_\mu \bar{u}_i \gamma^\mu u_j$	vector flavour-changing up-quark coupling to VMET	$A1FC_{13} = A1FC_{31} = 0.1$
B1FC[i, j]	$X_\mu \bar{u}_i \gamma^\mu \gamma^5 u_j$	axial-vector up-quark coupling to VMET	all zero
A12S[i]	$\phi^a \bar{u}_i^a \chi$	scalar coupling between phiC, up quark, and FMET	$A12S_3 = 0.1$
B12S[i]	$\phi^a \bar{u}_i^a \gamma^5 \chi$	pseudoscalar coupling between phiC, up quark, and FMET	all zero
tA12S[i]	$\tilde{\phi}^a \bar{d}_i^a \chi$	scalar coupling between tphiC, down quark, and FMET	$tA12S_1 = tA12S_2 = 0.1$
tB12S[i]	$\tilde{\phi}^a \bar{d}_i^a \gamma^5 \chi$	pseudoscalar coupling between tphiC, down quark, and FMET	all zero
AQS[i, j]	$\phi^c \epsilon_{abc} \bar{d}_i^{c,a} d_j^b$	scalar diquark coupling of phiC to two down quarks	$AQS_{12} = 0.1, AQS_{21} = -0.1$
BQS[i, j]	$\phi^c \epsilon_{abc} \bar{d}_i^{c,a} \gamma^5 d_j^b$	pseudoscalar diquark coupling of phiC to two down quarks	all zero
tAQS[i, j]	$\tilde{\phi}^c \epsilon_{abc} \bar{d}_i^{c,a} u_j^b$	scalar diquark coupling of tphiC to down-up quark pair	$tAQS_{13} = 0.1, tAQS_{23} = 0.1$
tBQS[i, j]	$\tilde{\phi}^c \epsilon_{abc} \bar{d}_i^{c,a} \gamma^5 u_j^b$	pseudoscalar diquark coupling of tphiC to down-up quark pair	all zero

Parameter	Multiplies	Physical meaning	Default nonzero entries
A12V[i]	$V_\mu^a \bar{u}_i^a \gamma^\mu \chi$	vector coupling between VC, up quark, and FMET	$A12V_3 = 0.1$
B12V[i]	$V_\mu^a \bar{u}_i^a \gamma^\mu \gamma^5 \chi$	axial-vector coupling between VC, up quark, and FMET	all zero
AQV[i, j]	$V_\mu^c \epsilon_{abc} \bar{d}_i^{c,a} \gamma^\mu d_j^b$	vector diquark coupling of VC to two down quarks	$AQV_{11} = 0.1$
BQV[i, j]	$V_\mu^c \epsilon_{abc} \bar{d}_i^{c,a} \gamma^\mu \gamma^5 d_j^b$	axial-vector diquark coupling of VC to two down quarks	all zero

Physics Summary

The file encodes a simplified extension of the SM with neutral missing-energy states: a real scalar SMET, real vector VMET, and self-conjugate fermion FMET, plus coloured scalar and vector mediators carrying $SU(3)_c$ triplet quantum numbers. The neutral scalar/vector couple flavour off-diagonally to up-type quarks, while the coloured mediators couple both to quark pairs through antisymmetric colour contractions and to a quark plus the neutral fermion. It mediates flavour-changing up-quark processes, diquark resonances, and quark-initiated production of invisible FMET states through coloured scalar or vector exchange.

Paper cross-check

Comparison of reconstruction.md Against the Paper Model

Located Paper Definitions

The relevant model definition is in the paper section “**EFFECTIVE THEORY FOR MONOTOPS**”. The field content is introduced immediately before Eq. (1): the paper denotes by ϕ, χ, V the scalar, fermionic, and vector missing-energy particles, and by φ, X colored scalar and vector fields in the fundamental representation of $SU(3)_c$. The complete effective interaction Lagrangian is given in **Eq. (1)**, followed by explanatory text specifying charge conjugation, color indices, flavor matrices, and the symmetry consequences of the ϵ_{ijk} tensor. Benchmark coupling choices are later stated in “**MODEL-INDEPENDENT SEARCHES**”, after the sentence beginning “To illustrate the main features...”.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Field definitions: ϕ, χ, V are scalar, fermionic, and vector missing-energy particles; φ, X are colored scalar/vector fields in fundamental $SU(3)_c$ representations, “EFFECTIVE THEORY FOR MONOTOPS” before Eq. (1).	$S = \text{SMET}, \chi = \text{FMET},$ $X_\mu = \text{VMET}, \phi^a = \text{phiC},$ $\tilde{\phi}^a = \text{tphiC}, V_\mu^a = \text{VC}.$	agree	Physics mapping is consistent, but notation is potentially confusing: reconstruction’s neutral vector X_μ corresponds to paper’s V_μ , while reconstruction’s colored vector V_μ^a corresponds to paper’s $X_{\mu,i}$.

paper term (eq. ref)	reconstruction term	verdict	notes
L_{SM} , Eq. (1).	Included only implicitly as $\mathcal{L}_{\text{LMono}}$ interactions; reconstruction focuses on BSM terms.	missing-in-reconstruction	Acceptable if reconstruction is only of the new model file interactions, but the paper's Eq. (1) explicitly includes L_{SM} .
$\phi \bar{u} [a_{FC}^0 + b_{FC}^0 \gamma^5] u$, Eq. (1).	$S \bar{u}_i^a [A0FC_{ij} + B0FC_{ij} \gamma^5] u_j^a + \text{h.c.}$	agree	Same neutral scalar flavor-changing up-quark coupling; reconstruction makes flavor/color indices explicit.
$V_\mu \bar{u} [a_{FC}^1 \gamma^\mu + b_{FC}^1 \gamma^\mu \gamma^5] u$, Eq. (1).	$X_\mu \bar{u}_i^a \gamma^\mu [A1FC_{ij} + B1FC_{ij} \gamma^5] u_j^a + \text{h.c.}$	agree	Same neutral vector flavor-changing up-quark coupling; $\gamma^\mu (A + B\gamma^5)$ is equivalent to $A\gamma^\mu + B\gamma^\mu \gamma^5$.
$\epsilon_{ijk} \varphi_i \bar{d}_j^c [a_{SR}^q + b_{SR}^q \gamma^5] d_k$, Eq. (1).	$\phi^c \epsilon_{abc} \bar{d}_i^{c,a} [AQ S_{ij} + BQ S_{ij} \gamma^5] d_j^b + \text{h.c.}$	agree	Same colored scalar diquark coupling to two down-type quarks with antisymmetric color tensor. Reconstruction uses separate color and generation indices.
$\varphi_i \bar{u}_i [a_{SR}^{1/2} + b_{SR}^{1/2} \gamma^5] \chi$, Eq. (1).	$\phi^a \bar{u}_i^a [A12S_i + B12S_i \gamma^5] \chi + \text{h.c.}$	agree	Same colored scalar coupling to an up-type quark and invisible fermion.
$\epsilon_{ijk} \tilde{\varphi}_i \bar{d}_j^c [\tilde{a}_{SR}^q + \tilde{b}_{SR}^q \gamma^5] u_k$, Eq. (1).	$\tilde{\phi}^c \epsilon_{abc} \bar{d}_i^{c,a} [tAQ S_{ij} + tBQ S_{ij} \gamma^5] u_j^b + \text{h.c.}$	agree	Same additional scalar coupling used for four-fermion interactions, with down-charge-conjugate/up bilinear.
$\tilde{\varphi}_i \bar{d}_i [\tilde{a}_{SR}^{1/2} + \tilde{b}_{SR}^{1/2} \gamma^5] \chi$, Eq. (1).	$\tilde{\phi}^a \bar{d}_i^a [tA12S_i + tB12S_i \gamma^5] \chi + \text{h.c.}$	agree	Same additional scalar coupling to a down-type quark and invisible fermion.
$\epsilon_{ijk} X_{\mu,i} \bar{d}_j^c [a_{VR}^q \gamma^\mu + b_{VR}^q \gamma^\mu \gamma^5] d_k$, Eq. (1).	$V_\mu^c \epsilon_{abc} \bar{d}_i^{c,a} \gamma^\mu [AQ V_{ij} + BQ V_{ij} \gamma^5] d_j^b + \text{h.c.}$	agree	Same colored vector diquark coupling. Reconstruction includes BQV , but its defaults set it to zero, matching the paper's later benchmark choice.
$X_{\mu,i} \bar{u}_i [a_{VR}^{1/2} \gamma^\mu + b_{VR}^{1/2} \gamma^\mu \gamma^5] \chi$, Eq. (1).	$V_\mu^a \bar{u}_i^a \gamma^\mu [A12V_i + B12V_i \gamma^5] \chi + \text{h.c.}$	agree	Same colored vector coupling to an up-type quark and invisible fermion.
Eq. (1) ends with +h.c..	Reconstruction states all displayed interaction terms have HC[...].	agree	Hermitian conjugation is consistently included.

paper term (eq. ref)	reconstruction term	verdict	notes
Text after Eq. (1): c denotes charge conjugation; color indices are fundamental; flavor indices are understood.	Reconstruction defines $q^c = C\bar{q}^T$, $\bar{q}^c$, ϵ_{abc} , and explicit flavor/color sums.	agree	Conjugation and index structure are compatible.
Text after Eq. (1): identical-quark couplings to scalar φ vanish, and axial couplings to vector X vanish, due to ϵ_{ijk} symmetry.	Reconstruction parameter defaults include $AQS_{12} = 0.1$, $AQS_{21} = -0.1$, $AQV_{11} = 0.1$, all $BQV = 0$.	agree	The benchmark choices respect the stated scalar antisymmetry and vanishing vector axial benchmark coupling.
Benchmark after Eq. (1): all axial couplings involving new particles vanish, $b = \tilde{b} = 0$.	Reconstruction defaults set all B -type couplings to zero.	agree	Matches the paper's illustrative scenarios.
Benchmark after Eq. (1): $(a_{FC}^0)_{13} = (a_{FC}^0)_{31} = (a_{FC}^1)_{13} = (a_{FC}^1)_{31} = (a_{SR}^q)_{12} = -(a_{SR}^q)_{21} = (a_{SR}^{1/2})_3 = (a_{VR}^q)_{11} = (a_{VR}^{1/2})_3 = a = 0.1$.	Reconstruction defaults list $A0FC_{13} = A0FC_{31} = 0.1$, $A1FC_{13} = A1FC_{31} = 0.1$, $AQS_{12} = 0.1$, $AQS_{21} = -0.1$, $A12S_3 = 0.1$, $AQV_{11} = 0.1$, $A12V_3 = 0.1$.	agree	Coupling pattern matches the paper's nonzero benchmark entries.
Four-fermion benchmark: $(a_{SR}^q)_{12} = -(a_{SR}^q)_{21} = (\tilde{a}_{SR}^q)_{\{1,2\}3} = (\tilde{a}_{SR}^{1/2})_{\{1,2\}} = a = 0.1$.	Reconstruction defaults list $tAQS_{13} = tAQS_{23} = 0.1$, $tA12S_1 = tA12S_2 = 0.1$.	agree	Same four-fermion-limit nonzero entries.
Paper does not explicitly assign $SU(2)_L$ representations or hypercharges in Eq. (1); it works in mass eigenstates.	Reconstruction states all new states are $SU(2)_L$ singlets and assigns $Y = 0, 2/3, -1/3$ as appropriate.	extra-in-reconstruction	These assignments are consistent with electric charge conservation under a singlet assumption, but they are not explicitly specified by the paper's effective Lagrangian.
Paper treats χ as a possible invisible fermionic particle and discusses massive or massless fermionic missing energy; it does not require Majorana self-conjugacy in Eq. (1).	Reconstruction says FMET is "self-conjugate, Majorana-like."	extra-in-reconstruction	This is a stronger field-property assumption than the paper states.
Paper treats ϕ and V as neutral missing-energy bosonic states, without explicitly declaring real versus complex fields in Eq. (1).	Reconstruction says SMET is a real scalar and VMET is a real vector.	extra-in-reconstruction	Reasonable for neutral invisible particles, but not fixed by the paper text shown in Eq. (1).

paper term (eq. ref)	reconstruction term	verdict	notes
Paper benchmark masses: resonant φ , X mass fixed to 500 GeV; four-fermion heavy masses $M_\varphi = M_{\tilde{\varphi}} = 3$ TeV; invisible masses vary by scenario.	Reconstruction field table lists defaults such as $M_\chi = 50$, $M_{V_{\text{MET}}} = 50$, $M_{VC} = 500$, $M_{\phi C} = 1000$, $M_{\tilde{\phi}C} = 1000$.	disagree	The interaction Lagrangian agrees, but the reconstruction's listed default masses do not reproduce all paper benchmark mass choices.

Disagreements and Human Checks

1. L_{SM} **omission** — **severity: convention**. A human should check whether `reconstruction.md` intentionally reconstructs only the BSM FeynRules interactions or whether the full model file should include/import the Standard Model Lagrangian.
2. **Extra electroweak quantum numbers** — **severity: convention**. A human should check the implementation file's actual class declarations to confirm whether $SU(2)_L$ singlet and hypercharge assignments were explicit model choices or inferred by the reconstruction.
3. **Self-conjugate invisible fermion χ** — **severity: substantive**. A human should check whether FMET is declared self-conjugate in the implementation, because the paper's Eq. (1) does not require χ to be Majorana-like.
4. **Real neutral scalar/vector assumptions** — **severity: convention**. A human should check whether SMET and VMET are explicitly self-conjugate in the implementation, since the paper only identifies them as neutral invisible bosonic states.
5. **Default mass values versus paper scenarios** — **severity: substantive for phenomenology, not for the interaction Lagrangian**. A human should check whether the reconstruction is reporting arbitrary implementation defaults or the benchmark masses used for the paper's five scenarios.
6. **Notation swap V/X** — **severity: cosmetic**. A human should check any downstream comparison carefully because the reconstruction uses X_μ for the paper's neutral vector V_μ , and V_μ^a for the paper's colored vector $X_{\mu,i}$.

Overall Assessment

The reconstructed interaction Lagrangian matches the physics content of the paper's Eq. (1) term by term: the neutral scalar/vector flavor-changing up-quark interactions, the colored scalar and vector diquark couplings, the quark- χ mediator couplings, the tilde-scalar four-fermion-limit interactions, hermitian conjugation, charge-conjugated down-quark structure, color antisymmetry, and the benchmark nonzero coupling pattern are all represented consistently. The main caveats are outside the core interaction terms: the reconstruction adds electroweak representation, hypercharge, self-conjugacy, and default mass information that the paper either leaves unspecified or varies by scenario.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.

6. Sanitizer scope: 22 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: